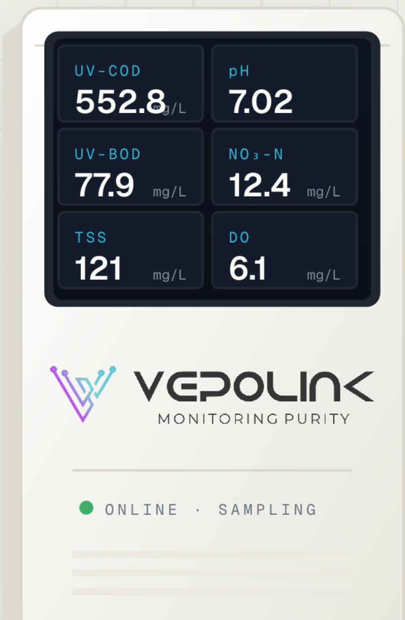
VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × WATER QUALITY

ONLINE WATER ANALYZER

Multi-Parameter *Optical* Water Analyzer

One instrument, up to 20 water-quality parameters — reagent-free optical measurement for continuous, real-time monitoring.



180–750 nm

UV-VIS SCANNING RANGE

7"

TOUCH-SCREEN INTERFACE

±5% FS

MEASUREMENT ACCURACY

RS485

MODBUS · REAL-TIME OUTPUT

180-750 NM UV-VIS OPTICS | **REAGENT-FREE** OPERATION | **IP 65** ENCLOSURE

20	7	±5%	0
PARAMETERS CONFIGURABLE	MEASURED SIMULTANEOUSLY	FS ACCURACY	REAGENTS REQUIRED

01 OVERVIEW

One analyzer for your *whole* water profile.

A high-performance online water analyzer that delivers accurate, reliable measurement of up to seven parameters at once — across a broad set of water-quality indicators — using state-of-the-art optical technology.

Built for exceptional stability, low operating cost, and high precision, it suits demanding environments from wastewater treatment plants to river monitoring stations. The modular platform lets you pick any of 20 parameters of choice, so a single instrument grows with your site.

MEASURABLE INDICATORS

- COD
- BOD
- TSS
- TOC
- TURBIDITY
- NITRATE
- NITRITE
- AMMONIUM
- TOTAL N
- TOTAL P
- DO
- CONDUCTIVITY
- PH
- COLOR
- OIL IN WATER

· PRODUCT FEATURES

Multi-Parameter Monitoring

Simultaneous measurement of organic matter, nitrate, color, turbidity and more — from one flow cell.

Advanced Optics

High-resolution spectrograph with a scanning wavelength range from 180 to 750 nm.

Durability & Auto-Cleaning

Built-in automatic cleaning system with long-lasting light source keeps optics stable over time.

Modular Design

Expandable with external probes for pH, conductivity, dissolved oxygen and specialist sensors.

02 MEASURED PARAMETERS

Parameters & *measuring ranges.*

UV SPECTROSCOPY — INTERNAL OPTICAL CELL

UV-COD	0 … 1000 mg/L*
UV-TSS	0 … 1000 mg/L*
UV-BOD	0 … 1000 mg/L*
UV-TOC	0 … 1000 mg/L*
Nitrite NO_2^-	0 … 100 mg/L*

NEPHELOMETRY — IR LIGHT SCATTER

Total Suspended Solids TSS	0.1 … 10000 mg/L*
Turbidity Turb	0.1 … 100 NTU*

VISIBLE FLUORESCENCE

Dissolved Oxygen DO	0.1 … 20 mg/L*

CONDUCTIMETER · POTENTIOMETRY

Conductivity EC	0.1 … 1000 mS/cm*
pH H^+	0 … 14*

ION SELECTIVE ELECTRODE (ISE)

Ammoniacal Nitrogen NH_4^+-N	0 … 3000 mg/L*
Total Phosphate TP	0 … 100 mg/L*
Nitrate Nitrogen NO_3-N	0 … 3000 mg/L*

* Ranges up to 3500 available on request. · Total Nitrogen (TN) is derived from Ammonia (NH_3-N), Nitrate (NO_3-N) and Nitrite (NO_2-N).

03 HOW IT WORKS

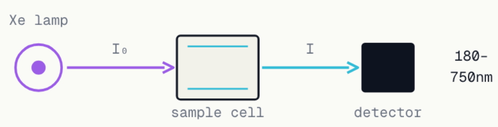
Four proven *measurement* principles.

UV spectroscopy enables rapid, reagent-free analysis of key water-quality parameters. Combined with three complementary sensing methods, the analyzer covers organics, solids, nutrients, oxygen and pH from a single platform.

UV-VIS SPECTROSCOPY

Absorbance — Beer-Lambert Law

COD · BOD · TOC · TSS · NITRATE · COLOR



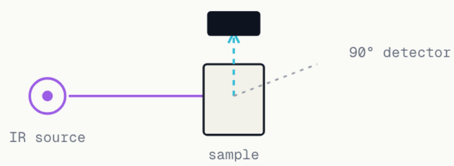
A high-resolution spectrograph measures the difference between incident light on the reference and light transmitted through the sample. Concentration is linear with absorbance and optical path length.

$A = \log(I_0 / I)$

NEPHELOMETRY

IR Light Scatter — ISO 7027

TURBIDITY · TSS



An infrared beam passes through the sample and scatters. A detector converts scattered-light intensity into a signal — the turbidity value rises with higher scattered intensity.

`90° scattered detection`

VISIBLE FLUORESCENCE

Oxygen Quenching

DISSOLVED OXYGEN

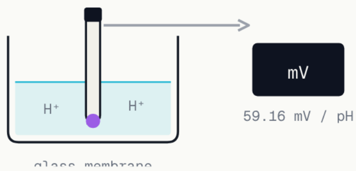
A blue LED excites a fluorescent layer; oxygen passing through the DO-permeable membrane quenches the fluorescence. Higher DO means stronger quenching and less detected light.

$DO \propto 1 / \text{fluorescence}$

POTENTIOMETRY

Nernst Equation

PH



pH is the potential difference between reference and sensing half-cells across a glass membrane. At 25 °C, one pH unit equals 59.16 mV; pH 7 outputs 0 V at all temperatures.

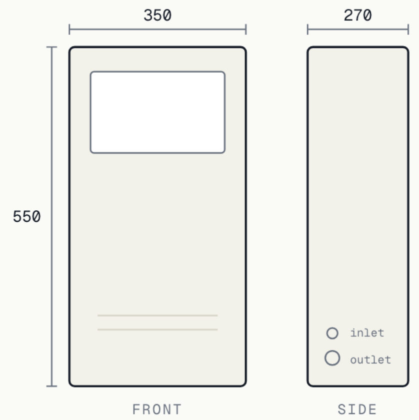
`1 pH unit = 59.16 mV @ 25 °C`

04 SPECIFICATIONS

Technical *specifications.*

Display	7-inch touch screen
Power Supply	85-500 VAC / 9-36 VDC
Power Consumption	40 VA · 25 W max
Analog Output	4-20 mA (optional)
Relay	Programmable response
Communication	MODBUS RS485 · USB opt.
Light Source	Xenon flash lamp
Accuracy	±5% FS
Optical Path Length	1 – 65 mm selectable
Operating Temperature	0 – 50 °C
Storage Temperature	-10 – 65 °C
Enclosure Rating	IP 65
Dimensions (L×W×H)	350 × 270 × 550 mm
Cleaning	Reagent-free · DM water
Fittings	Inlet ⌀8 · Outlet ⌀12

ENCLOSURE DIMENSIONS



FRONT & SIDE ELEVATION · ALL VALUES IN MM

05 MODULAR PROBES

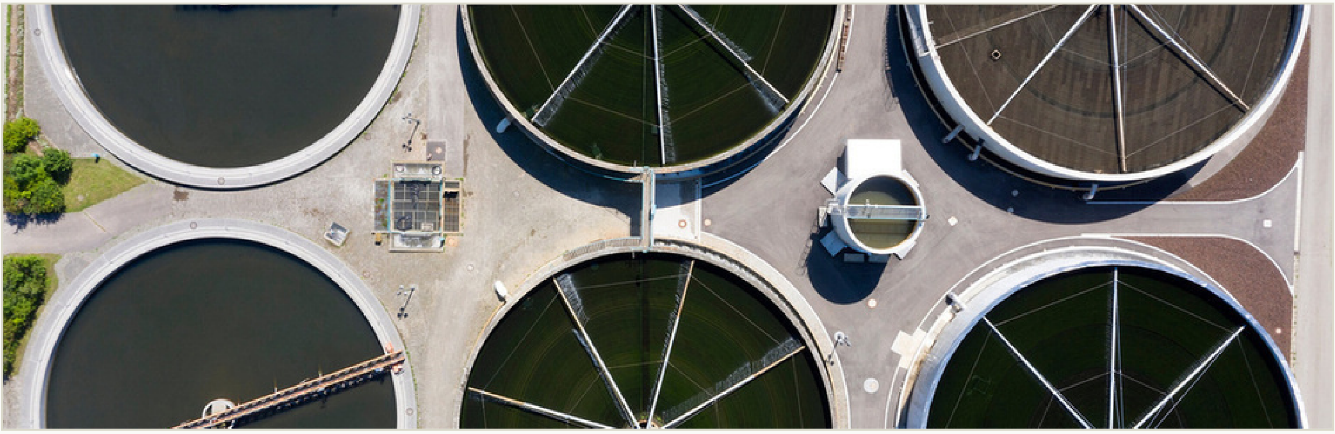
Compatible *sensors* & ordering codes.

Expand the analyzer with digital and analog probes. Standard cable lengths ship with each sensor and can be extended on request.

ORDERING CODE	DESCRIPTION
VS-TurbLo	**Low Turbidity Sensor** 0.001-100 NTU · 3 m std (ext. 30 m)
VS-Turb	**Optical Turbidity Sensor** 0.01-4000 NTU · 10 m std (ext. 100 m)
VS-SS	**Suspended Solids / Sludge Sensor** 0.01-20000 mg/L · 10 m std (ext. 100 m)
VS-NO3N	**Nitrate Nitrogen Sensor** 0-3000 mg/L NO_3N · 5 m std (ext. 100 m)
VS-NH4N	**Ammonia Nitrogen Sensor** 0.1-3000 mg/L NH_4N · 5 m std (ext. 100 m)
VS-RCL	**Residual Chlorine Sensor** 0-20 mg/L · 10 m std (ext. 100 m)
VS-ODO	**Optical DO Sensor** 0-20 mg/L / 0-200% sat. · 10 m std
VS-OIW	**Oil in Water Sensor** 0-50 ppm · 10 m std (ext. 50 m)
VS-BGA	**Blue-Green Algae Sensor** 0-300000 cells/mL · 10 m std
VS-CPHL	**Chlorophyll Sensor** 0-500 µg/L · 10 m std (ext. 100 m)
VS-EC	**Conductivity Sensor — Digital / Analog** 0-20 mS/cm · 10 m std
VS-pH	**pH Sensor — Digital / Analog** 0-14 pH · 10 m std
VS-ORP	**ORP Sensor — Analog** ±2000 mV · 10 m std (ext. 20 m)
VS-TDS	**TDS Sensor — Analog** 0-20000 ppm · 10 m std (ext. 20 m)

06 APPLICATIONS

Where it *works.*



Wastewater Treatment Plants	River Monitoring Stations	Potable Water Treatment
Desalination Plants	Chlor-Alkali Industry	Fish Farming & Aquaculture
Food & Beverage	Industrial Process Water	Chemical Processing

07 FAQ

Frequently asked *questions.*

Q How many parameters can it measure at once?

Up to seven parameters simultaneously. The modular platform lets you configure any of 20 parameters of choice, so one instrument can cover your full monitoring profile.

Q How does it connect to my SCADA / control system?

MODBUS RS485 is standard and transmits measured values in real time. A 4–20 mA analog output, programmable relay and USB port are available as options.

Q Does it need reagents or chemicals?

No. Core measurement is fully reagent-free, based on optical UV-Vis spectroscopy. Only DM water is required for the automatic cleaning cycle — keeping operating costs low.

Q Is it suitable for outdoor or harsh installations?

The enclosure is rated IP 65 and operates from 0–50 °C, with storage down to −10 °C — suitable for demanding plant and field environments.

Q How does it measure without chemicals?

It uses UV-Vis absorbance (Beer-Lambert law), IR nephelometry, visible fluorescence and potentiometry — reading the water's optical and electrochemical response directly.

Q How much maintenance does it need?

Minimal. The built-in automatic cleaning system and long-life lamp reduce manual upkeep. The 7-inch touch screen includes maintenance diagnostics for quick checks.

Q What is the measurement accuracy?

±5% of full scale, with a high-resolution spectrograph and Xenon flash light source engineered for long-term stability.

Q Can I measure Total Nitrogen?

Yes — Total Nitrogen (TN) is derived from Ammonia (NH_3-N), Nitrate (NO_3-N) and Nitrite (NO_2-N) measured by the analyzer.

Q Can I add sensors later?

Yes. The analyzer is modular and expandable with external probes — pH, conductivity, DO, turbidity, chlorine, algae and more — using standard ordering codes.

Q What sample connections are required?

A straight-through hard-pipe inlet (⌀8) and outlet (⌀12). Optical path length is selectable from 1 to 65 mm to match your application and concentration range.

This brochure is general product information. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

MONITORING PURITY, IN REAL TIME

See every parameter. *Continuously.*

Tell us which parameters your site needs to track and we'll configure the right analyzer — with the sensors, ranges and integration to match. Pick how you'd like to start.

01

Talk to our engineers

Speak directly to the team that specifies and commissions water analyzers. No call centre.

+91 99538 86178 →

02

Request a parameter configuration

RECOMMENDED

Send us your parameters and ranges — we'll spec the analyzer, sensors and outputs for your site.

sales@vepolink.com →

03

Book a demonstration

See the analyzer measure your parameters, live, with our engineering team.

Request a demo →